

```

disabledatascaling,
...
]
\addplot3[
    mark=*,mark size=3pt,only marks,draw=black,fill=gray,
    nodes near coords=\coordindex,
    every node near coord/.append style={yshift=3pt}
]
table {
X      Y      Z
0.324  0.213  0.463
...    ...    ...
0.150  0.700  0.150
};
\end{ternaryaxis}

```

An easy technique for numbering each point is to use *coordindex*. See Reference 13, pages 487–498, in which more complex examples are reported.

Flowchart

This section is dedicated to creating a basic example of a flowchart with the package *tikz*. Even in this case, LaTeX can produce a great output. In the preamble, the packages *xcolor*, *mhchem* (because some chemical formulae will be used) and *tikz* are called. Then each chart component (block, decision — a type of diamond shaped block, and line) is defined using *tikzstyle* — for example, geometry, fill colour, text, etc. In the *tikzpicture* section, each block or decision (both called *node*) is positioned with respect to the others and the text content is written. Last, every node is connected with a *path* (a line), and on each path it's also possible to add some text. It's pretty easy and only takes a little patience. Probably, the best explanation is to view a practical example (Figure 16):

```

\documentclass[border=0.5cm]{standalone}
\usepackage[svgnames]{xcolor}
\usepackage[version=3]{mhchem}
\usepackage{tikz}
\usetikzlibrary{shapes,arrows}
\tikzstyle{block1}=[rectangle,draw,fill=Tomato,text
width=2cm,minimum height=2cm,rounded corners]
...
\tikzstyle{line}=[draw,-triangle 45]
\begin{document}
\begin{tikzpicture}[text centered,node distance=4cm,auto]
\node [block1]          (1) {\text{Mixture}};
\node [decision,right of=1] (2) {\text{+}\ce{CHCl3}};

```

```

...
\path [line] (1)--(2);
\path [line] (2)--(3);
...
\path [line] (2)--node {Aq} (3);
\path [line] (2)--node {Org} (5);
...
\end{tikzpicture}
\end{document}

```

Only some introductory examples are presented here. More of them can be found in References 11 and 12, and a large collection in the manuals for the packages *pgfplots* and *pst-solides3d* (References 13 and 14). About *tikz*, an introduction can be found in Reference 15 and a large manual in Reference 16. 

References

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